1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

CA-125 (MUC16) participates in a mesothelin-dependent interaction network that drives adhesion and shedding across ovarian, mesothelioma, and select gastrointestinal cancers, producing elevated serum CA-125 in those contexts. In prostate cancer, CA-125 elevation is rare and not tightly coupled to tumor burden, indicating limited disease-specific utility. Consequently, CA-125 lacks specificity for prostate cancer unless combined with mesothelin pathway activation markers. A multi-analyte diagnostic approach incorporating CA-125/MUC16, mesothelin (MSLN or soluble mesothelin-related peptide), and PSA will improve discrimination between prostate cancer and non-prostate, mesothelin-associated diseases. Testable predictions: (1) in cancers with high MSLN expression, serum CA-125 correlates with MUC16–mesothelin axis activity and not with prostate cancer stage; (2) disruption of the MUC16–mesothelin interaction reduces adhesion, migration, or metastasis in mesothelin-expressing tumor models; (3) a combined CA-125/PSA/soluble mesothelin panel provides superior diagnostic accuracy for distinguishing prostate-origin disease from ovarian/mesothelin-driven diseases in patients with elevated CA-125.

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: is a well-established biomarker for ovarian cancer but is elevated in a variety of benign and malignant conditions, limiting its disease-specificity. MUC16 physically interacts with mesothelin, a pairing implicated in tumor cell adhesion and metastatic processes in mesothelin-expressing cancers such as ovarian cancer and mesothelioma.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions